Department of Electrical, Electronic and Computer Engineering

Assessment ID: 2021EBN111A05

Electricity and Electronics EBN 111

Release date: 12 May 2021

Marks:	25	Total number of pages:	11
Submission Deadline:	25 May 2021 (23:59)	Open/Closed book:	Open

Important instructions

1. This PDF file must be used in together with the **Excel answer spreadsheet** found on your AMS system account. The **Excel answer spreadsheet** offers each of you the unique parameter values for each question. Some extra instructions are also available on the **Excel answer spreadsheet**.
2. Final answers must be written as real numbers and not as fractions. (Use at least 5 significant figures to write irrational numbers.)
3. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the **Excel answer spreadsheet**.)
4. You can upload the **Excel answer spreadsheet** to AMS multiple times before the submission deadline. You can self-grade two attempts of these uploads to check and possibly improve your answers. The mark for each assignment is awarded according to the final submission of the **Excel answer spreadsheet**.

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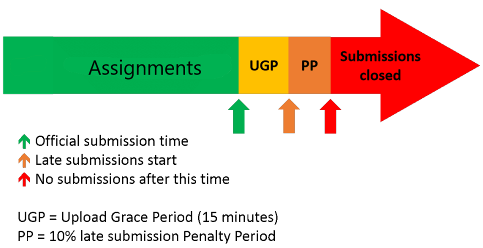
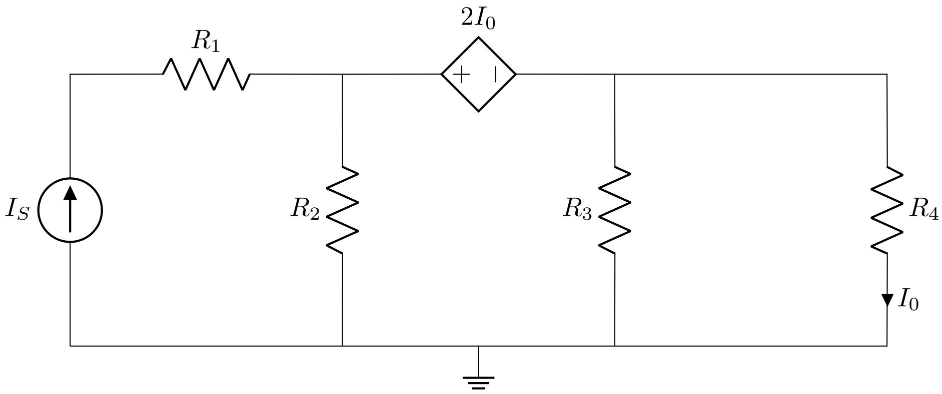
Assistant Lecturers: M. Maritz, J. Masela, J. Thiselton, M. Trivedi, D. Wepener

Online assessment submission rules and time-line

The following rules and time-line apply for all Assignments for EBN111/122 as illustrated by the figure below:

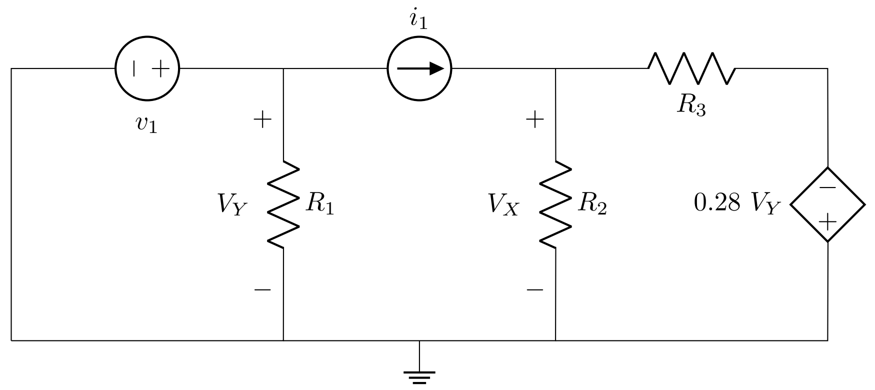
1. The start of the green section in the figure indicates the release of an assignment. The release of the assignment will be announced on ClickUP.
2. The official due date for the assignment is indicated by the green arrow in the figure. The due dates are also shown on the Lecture Schedule on ClickUP and on the AMS.
3. You must submit your **Excel answer spreadsheet** with your answers to each assignment on the AMS before the due date (green arrow in the figure below).
4. If you experience a technical issue with respect to uploading your assignment on the AMS, you can request assistance via **e-mail** ebn2021.ams@gmail.com. Note, except for extraordinary cases, you cannot submit your answers via email.
5. The submission due date will be followed by an Upload Grace Period of 15 minutes during which you must submit your answers on the AMS even if not finished with the assignment. No email submissions will be accepted during this period.
6. The Upload Grace Period is followed by a Penalty Period of 10 minutes. If you submit during this period, a 10% penalty will be applied to your final mark.
7. No late submissions will be accepted after the Penalty Period.



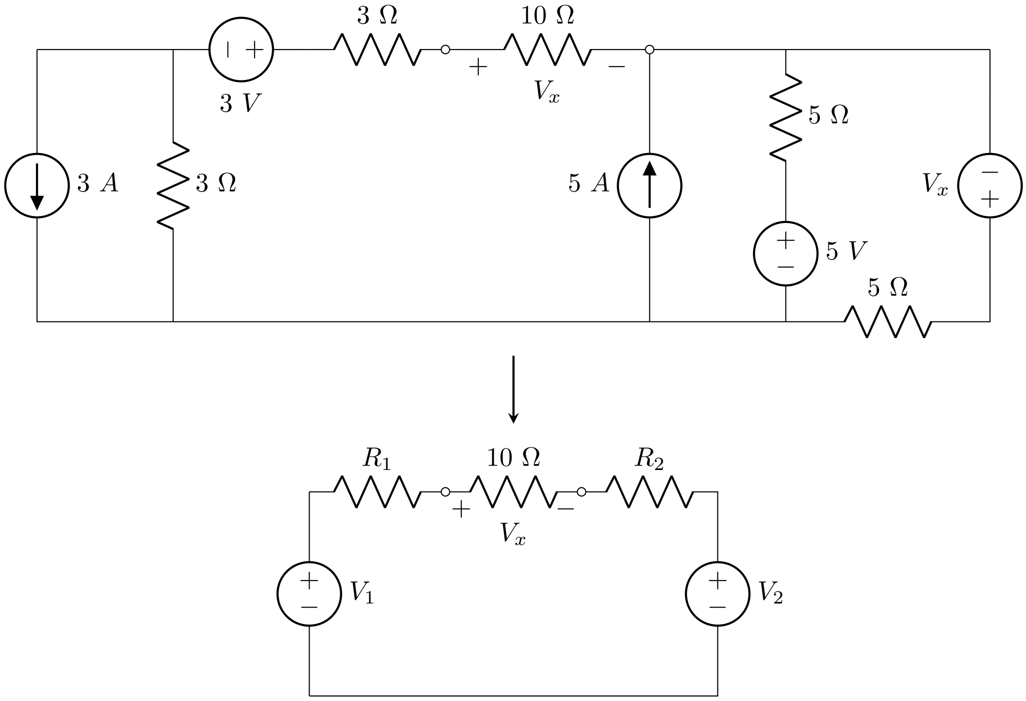
Question 1 [2]	
Use the principle of linearity to determine I_0/I_S, if R_1 = [R] Ω, R_2 = [S] Ω, R_3 = [T] Ω and R_4 = [Q] Ω	
	
01	I_0/I_S [2]

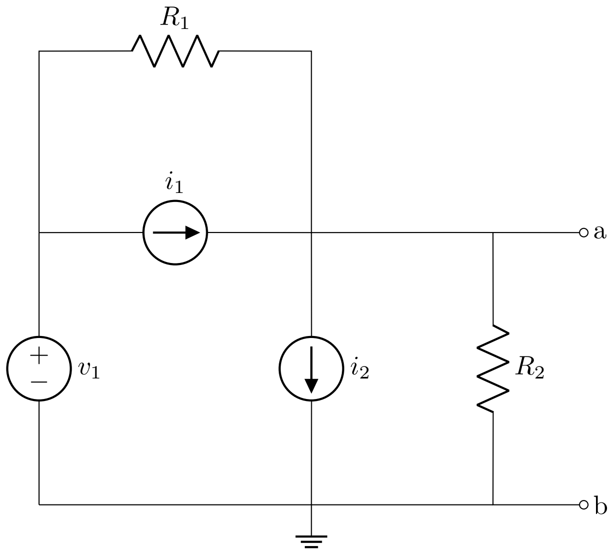
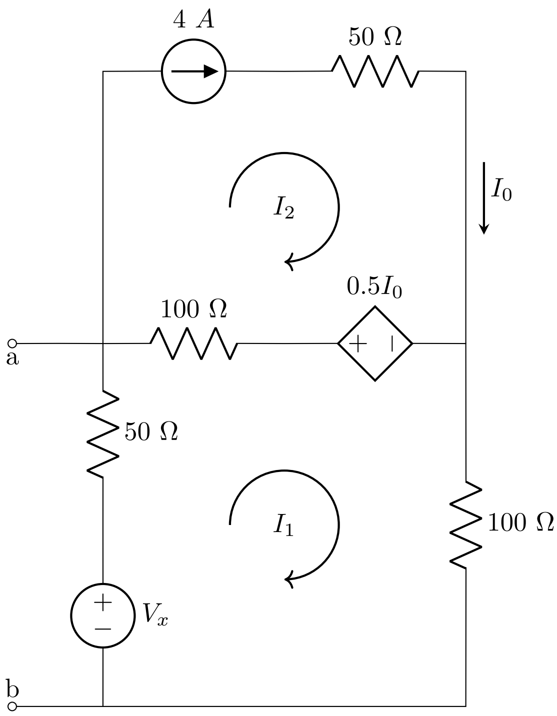
Questions 2 to 3 [4]

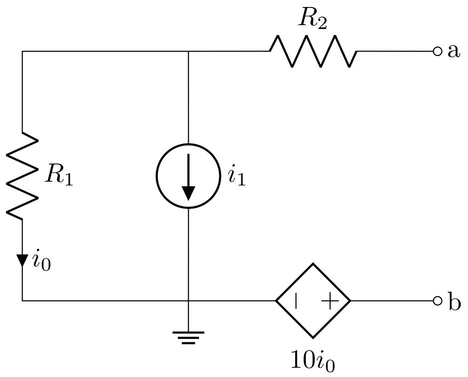
If R_1 = [R] Ω, R_2 = [S] Ω, R_3 = [T] Ω, v_1 = [Q] V and i_1 = [I] mA, determine the individual contribution of the:



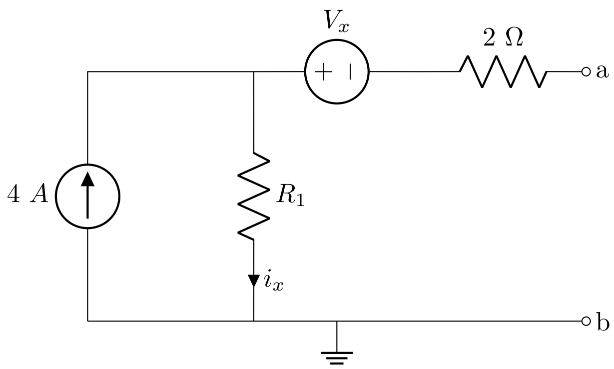
02	i_1 current source to V_x [2]
03	v_1 voltage source to V_x [2]

Question 4 [2]	
Use source transformation to simplify the circuit below for $V_x = [X]\ V$. $V_1 = -6\ V$, $R_1 = 6\ \Omega$ and $R_2 = 2.5\ \Omega$ are given, find V_2.	
	
04	V_2 [2]

Questions 5 to 6 [4]	
Find the Thevenin equivalent voltage V_{TH} with regards to terminals a-b. Let $R_1 = [R]\ \Omega$, $R_2 = [S]\ \Omega$, $v_1 = [V]\ V$, $i_1 = [Q]\ mA$ and $i_2 = [P]\ mA$.	
	
05	V_{TH} [2]
Find the Thevenin equivalent voltage V_{TH} between terminals a-b if $V_x = [X]\ V$.	
	
06	V_{TH} [2]

Questions 7 to 8 [4]	
Let $R_1 = [R]\ \Omega$, $R_2 = [S]\ \Omega$ and $i_1 = [I]\ A$. Determine the Norton equivalent current I_N and resistance R_N.	
	
07	I_N [2]
08	R_N [2]

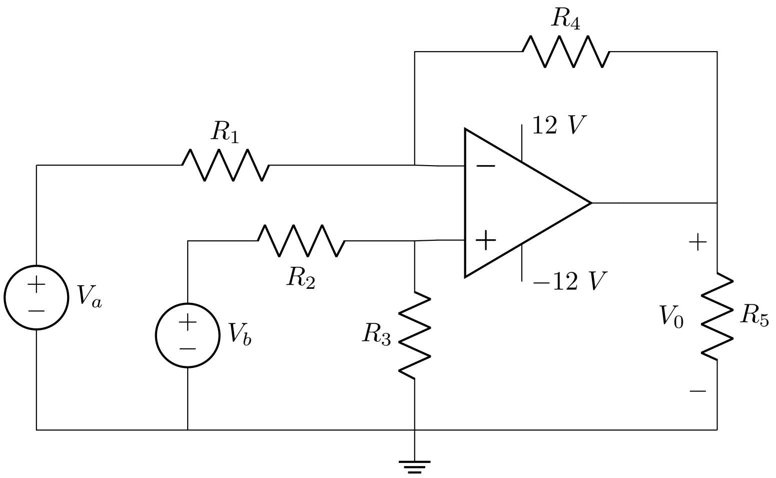
Questions 9 to 11 [4]	
Find the Thevinin equivalent voltage V_{TH} and resistance R_{TH}, and then calculate the maximum power that can be transferred to a load resistor R_L that can be connected between terminals a-b. $R_1 = [R]\ \Omega$ and $V_x = [V]\ V$.	



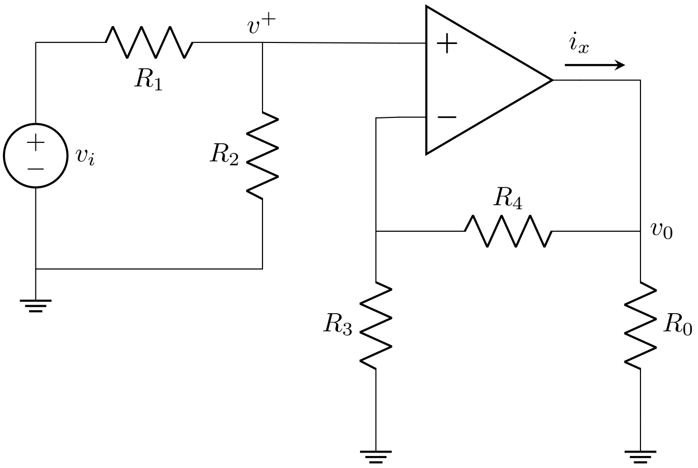
09	V_{TH} [2]
10	R_{TH} [1]
11	P_{max} [1]

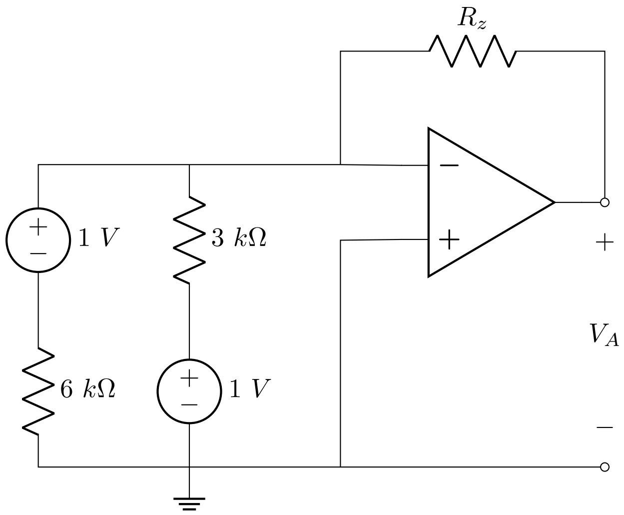
Question 12 [1]

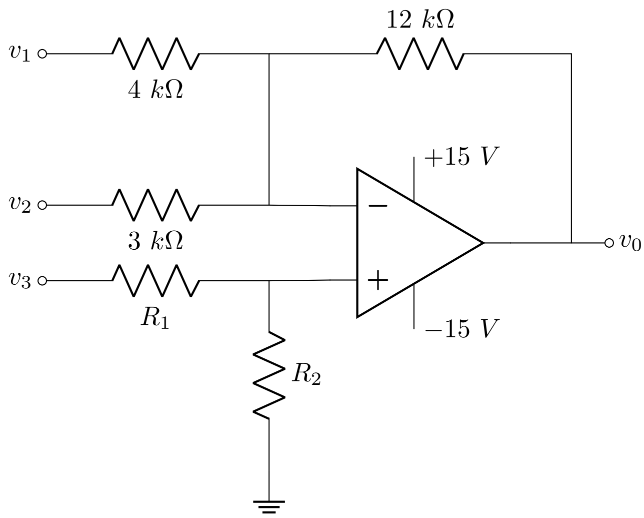
Find V_0 with $V_a = [V]\ V$, $V_b = [U]\ V$, $R_1 = [R]\ \Omega$, $R_2 = [S]\ \Omega$, $R_3 = [T]\ \Omega$, $R_4 = [P]\ \Omega$ and $R_5 = [Q]\ \Omega$.



12	V_0 [1]

Questions 13 to 14 [2]	
Let $R_3 = [A]\ k\Omega$ and $R_4 = [B]\ k\Omega$.	
	
13	Find the current i_x when $R_0 = [C]\ \Omega$ and $v^+ = -[D]\ V$. [1]
14	Calculate the closed-loop gain (v_0/v_i) of the circuit when $R_1 = R_2$. [1]

Question 15 [1]	
Given $R_z = [x]\ k\Omega$, determine V_A.	
	
15	V_A [1]

Question 16 [1]	
Find v_1 when $v_0 = -14\ V$, $v_2 = 2\ V$, $v_3 = [V]\ V$, $R_1 = [R]\ k\Omega$ and $R_2 = [S]\ k\Omega$	
	
16	v_1 [1]

Grand Total : [25]